

PHD Application Proposal:

Title:

AI adaptation in Digital Games: Psychological Mechanisms, Player Awareness and the Impact of AI Labelling on Player Experience.

Abstract:

Artificial Intelligence (AI) is increasingly used to personalise human experiences, from healthcare recommendations to educational platforms. Digital games represent one of the most advanced applications of this trend, with AI-driven systems that monitor player behaviour in real time and adjust game difficulty accordingly. These adaptive systems are designed to enhance player experiences by keeping challenge matched to skill, yet remarkably little is known about the psychological mechanisms through which this adaptation actually affects players. Even less is understood about what happens when players know AI is involved. Research outside gaming has shown that simply labelling something as AI-driven can change how people evaluate it, sometimes positively and sometimes negatively, regardless of whether the AI is actually doing anything different. One study in gaming found that merely telling players a game was adapting to them changed their reported experience, even when no adaptation was occurring. However, the psychological processes underlying this effect remain unexplored.

This project investigates three interconnected questions. First, what are the specific psychological mechanisms through which AI-driven game adaptation affects player experience? Second, does telling players that AI is adapting their game enhance or undermine those psychological effects? Third, can the AI label alone, without any actual AI, produce measurable changes in player experience? Using a controlled experiment in which both AI adaptation and what participants are told about AI are independently manipulated, this project will disentangle the effects of the technology itself from the effects of believing the technology is present. The findings will extend beyond gaming to inform how AI-driven personalisation is designed, communicated and regulated across society.

Section 1: Research Background

Artificial intelligence (AI) is increasingly embedded in the systems that shape everyday human experiences, from personalised content recommendations to adaptive learning platforms. Nowhere is this integration more apparent than in digital games, where AI-driven systems monitor player behaviour in real time and adjust difficulty, pacing, enemy behaviour and narrative pathways to suit individual players (Hunicke, 2005; Zohaib, 2018). As AI-personalised experiences become more prevalent across entertainment, education and beyond, understanding how these systems affect human psychology is not merely an academic question but a societal one.

The technical foundations of AI-driven game adaptation are well established. Dynamic difficulty adjustment (DDA) systems have been developed and refined over decades (Hunicke, 2005), and experience-driven procedural content generation extends adaptation beyond difficulty into broader design elements (Yannakakis & Togelius, 2011; Pfau et al., 2020). However, this literature is overwhelmingly technical in focus, prioritising system design over psychological understanding. Where player experience is considered, it is typically treated as a design outcome to be optimised rather than a psychological process to be understood (Zohaib, 2018). The specific psychological mechanisms through which AI adaptation influences player experience remain largely unexamined.

Two established frameworks address this gap. Self-Determination Theory (SDT; Deci & Ryan, 1985, 2000) proposes that wellbeing and intrinsic motivation depend on satisfying three basic needs: autonomy, competence and relatedness. The Player Experience of Need Satisfaction (PENS) model demonstrates that games fulfilling these needs produce deeper engagement (Ryan, Rigby & Przybylski, 2006; Przybylski et al., 2010). Flow Theory (Csikszentmihalyi, 1990) identifies the absorption that arises when perceived challenge matches perceived skill (Nakamura & Csikszentmihalyi, 2002). These frameworks are complementary components of a single process: AI-driven adaptation manipulates the challenge-skill balance underlying flow, and when that balance is achieved it simultaneously satisfies competence. Both depend on perceived autonomy. If awareness of AI involvement shifts that perception, it could disrupt the need satisfaction and flow the adaptation was designed to support, even when adaptation is objectively effective. Yet neither framework has been systematically applied in AI-adaptive gaming contexts.

A further dimension complicates this picture. A growing body of research demonstrates that simply knowing AI is involved in a process can alter how people evaluate it. This “algorithmic aversion” can lead people to reject AI-driven outcomes even when they match or exceed human-generated alternatives (Dietvorst, Simmons & Massey, 2015; Bigman & Gray, 2018; Burton et al., 2020), although “algorithm appreciation” has also been documented (Logg, Minson & Moore, 2019; Castelo, Bos & Lehmann, 2019). The most directly relevant empirical work in this space is Denisova and Cairns (2015), who demonstrated that merely telling players a game was adapting to them changed their reported experience, regardless of whether adaptation was occurring.

This finding establishes that player perception of AI can be experimentally separated from actual AI functionality. What remains unexamined is why this effect occurs: the specific psychological mechanisms driving it, whether the effect can operate negatively as well as positively, and how individual differences shape player responses. This project addresses these interconnected gaps through a single experimental framework, bringing together psychological theory, AI perception research and games research to examine not only what AI adaptation does to players, but what *believing* in AI adaptation does.

Section 2: Fit to Theme

This project sits within the Transformative Technologies for Society theme. AI has moved from a specialist tool to a pervasive societal force across healthcare, education, finance, media, and public services. Yet a fundamental psychological question remains underexamined: how do people actually react to AI-driven adaptation, and does knowing AI is involved change its impact? Games offer a unique methodological advantage - they allow real-time AI adaptation to be implemented, manipulated and measured within an ecologically valid interactive experience. The findings, however, extend beyond gaming. By identifying the psychological mechanisms through which AI adaptation affects experience and testing whether awareness of AI involvement alters those effects, this research produces insights that are directly transferable to any domain where AI personalisation shapes human outcomes. The project straddles the ESRC (Psychology - understanding human experience and behaviour) and EPSRC (Computing/AI - the adaptive systems being investigated), requiring substantive engagement with both disciplines to answer its core questions.

Section 3: Research Questions and aims

This project asks: how does AI-driven adaptation in digital games affect player psychological experience, and to what extent does player awareness of AI involvement moderate these effects?

RQ1: What are the psychological mechanisms - specifically need satisfaction (autonomy, competence) and flow - through which AI-driven game adaptation affects player experience?

RQ2: Does player awareness that AI is adapting their experience alter the psychological impact of the adaptation?

RQ3: Can the AI label alone, independent of actual AI adaptation, produce measurable placebo or nocebo effects on player psychological experience?

Individual differences in AI attitudes and gaming experience will be examined as primary moderators of the effects investigated in RQ2 and RQ3; these represent the individual-level

factors most directly implicated by the theoretical framework. Desirability of control will be measured as a covariate to account for its potential influence on perceived autonomy.

Research Aims:

Aim 1: To identify and measure the specific psychological mechanisms through which AI-driven game adaptation affects player experience, examining how adaptation influences the challenge-skill balance central to flow and the need satisfaction (autonomy, competence) central to SDT.

Aim 2: To investigate whether transparency about AI adaptation enhances or undermines player psychological outcomes.

Aim 3: To test whether AI labelling produces placebo or nocebo effects on player experience independent of actual adaptation, and to examine the role of the machine heuristic (Sundar, 2020) in driving these effects.

Section 4: Research Plan and Methodology

This project employs a 2x2 between-subjects factorial design. The two independent variables are actual AI adaptation (Present vs. Absent) and AI label (Told AI is adapting vs. Told AI is not adapting). Participants will be randomly allocated to one of four conditions and will complete the study once. Between-subjects design is essential – participants cannot be ‘untold’ about AI involvement.

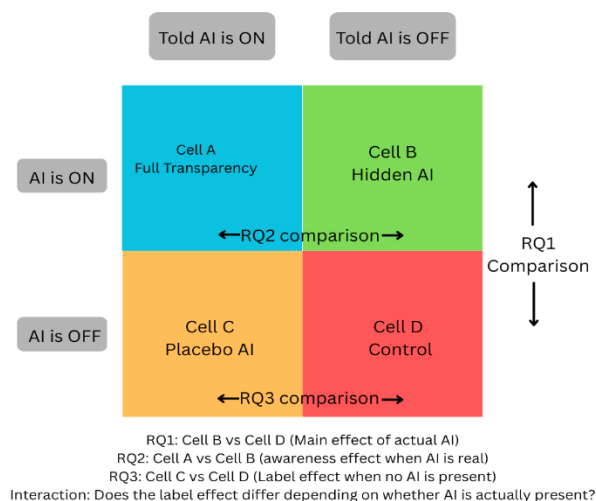


Figure 1. Experimental design and condition mapping. Rows represent actual AI status; columns represent participant information. Arrows indicate primary comparisons for each research question.

Approximately 120 to 160 students (30 to 40 per cell) will be recruited from the University of Surrey participant pool, a sample size informed by conventions for detecting medium effect sizes in factorial designs. Participants will require basic familiarity with digital games but no prior experience with the specific game used, as familiarity would confound key outcome measures.

Participants will play an existing game selected for its ecological validity as an interactive experience. The game genre will be selected to naturally engage the psychological constructs under investigation: genres requiring continuous player-initiated action engage autonomy, while those providing immediate, unambiguous performance feedback engage competence, ensuring that AI-driven difficulty adjustments produce measurable variance in the target constructs. A rule-based adaptive system will be developed and integrated with the game environment, designed to mirror the dynamic difficulty adjustment algorithms used in commercial games (Hunicke, 2005). The system will monitor real-time player performance inputs such as failure rate, score trajectory and completion speed, and adjust difficulty outputs such as enemy behaviour, obstacle frequency and resource availability according to defined algorithmic thresholds. This constitutes a genuine implementation of industry DDA while providing full experimental control.

In AI-present conditions, the system actively adjusts difficulty in response to player performance (addressing Aim 1); in AI-absent conditions, it is inactive and difficulty fixed at baseline. All other aspects, including visuals, mechanics and session structure will be held constant.

The factorial structure disentangles adaptation effects from the effects of believing adaptation is present (addressing Aims 2 and 3). Development of the adaptive system will be supported by the secondary supervisors' AI and computing expertise.

Participants will provide informed consent before completing pre-task measures assessing AI attitudes (General Attitudes toward AI scale; Schepman & Rodway, 2020), gaming experience and desirability of control (Burger & Cooper, 1979). They will then receive a standardised briefing indicating that the game either uses or does not use AI-driven difficulty adaptation. Participants will play the game environment for approximately 20 minutes. During gameplay, objective behavioural data will be logged, including in-game performance, behavioural persistence after failure, and decision patterns. System logs will verify that the adaptive algorithm adjusted difficulty parameters as intended in AI-present conditions and maintained baseline values in AI-absent conditions. Following gameplay, participants will complete the Player Experience of Need Satisfaction scale (PENS; Ryan, Rigby & Przybylski, 2006) as the primary outcome, capturing the need satisfaction mechanisms through which adaptation is theorised to affect experience. The Flow Short Scale (FSS-2; Engeser & Rheinberg, 2008) will serve as a secondary outcome, capturing the experiential state that need satisfaction supports. A manipulation check and items assessing perceived personalisation will be collected. Full debriefing will follow. As this study involves deliberate deception in two conditions, ethical approval will be sought in accordance with the ESRC Framework for Research Ethics.

Data will be analysed using factorial ANOVA and MANOVA to test main effects of actual AI adaptation, main effects of AI labelling and their interaction across outcome measures. Where pre-task measures differ significantly between groups, ANCOVA will be employed. Moderation

regression will examine whether AI attitudes and gaming experience moderate the effects of AI labelling on psychological outcomes; these are prioritised as the moderators most directly implicated by the theoretical framework, while desirability of control will be included as a covariate. A subsequent online replication phase will test the generalisability of these findings with a larger and more demographically diverse sample.

Section 5: Potential impact

This project bridges three bodies of literature that have developed largely in isolation: AI-driven game adaptation, psychological theories of player experience, and AI perception research. By testing these mechanisms and whether AI awareness moderates them, the findings will advance theoretical understanding across all three domains. Integrating SDT, Flow Theory and the machine heuristic into a single experimental paradigm will provide a model for future research on human responses to AI-adaptive systems.

For the games industry, the findings will directly inform design decisions about whether and how to disclose AI-driven adaptation to players. The required three-month placement with a game studio or AI company will facilitate direct knowledge exchange between research and industry practice.

Beyond gaming, AI-driven personalisation is embedded across healthcare, education, finance and public services. The psychological principles identified in this research, particularly whether the AI label alone can shift human experience, are transferable to those domains. This is timely given the growing policy landscape around AI transparency, including frameworks such as the EU AI Act that mandate users be informed when interacting with AI systems. This project provides empirical evidence to inform such policies by testing whether transparency about AI produces psychological benefits or costs. The paradigm is extensible: future phases may include online replication with more diverse samples, gamer/non-gamer comparison, or physiological measures and computational modelling to further examine these mechanisms.

Section 6: Proposed Timeline

Year 1 (months 1-12): A literature review will refine the framework and identify the optimal game environment. The adaptive system will be developed with the Computing supervisors, and all measures finalised. Ethical approval will be sought, with attention to the deception protocol. Pilot testing in the final months will validate the manipulation, test the system and confirm measures capture the intended constructs. Surrey's participant pool and laboratory facilities provide the infrastructure for this phase.

Year 2 (Months 13-24): Main data collection will take place, recruiting 120 to 160 participants across the four experimental conditions. Following data collection, the three-month placement will be undertaken with a game studio or AI company, facilitating knowledge exchange between research and industry. Initial data analysis will commence alongside the placement.

Year 3 and Beyond (Months 25 - 39): Full data analysis will be completed, including factorial ANOVA, MANOVA and moderation regression analyses. An online replication study will extend the primary findings to a broader and more diverse sample, including systematic comparison between experienced gamers and non-gamers. Findings from both phases will be prepared for publication in targeted journals and presented at relevant conferences such as CHI Play. The final months will be dedicated to thesis writing, revisions and submission.

Gantt chart showing a proposed PhD project timeline across 3.5 years

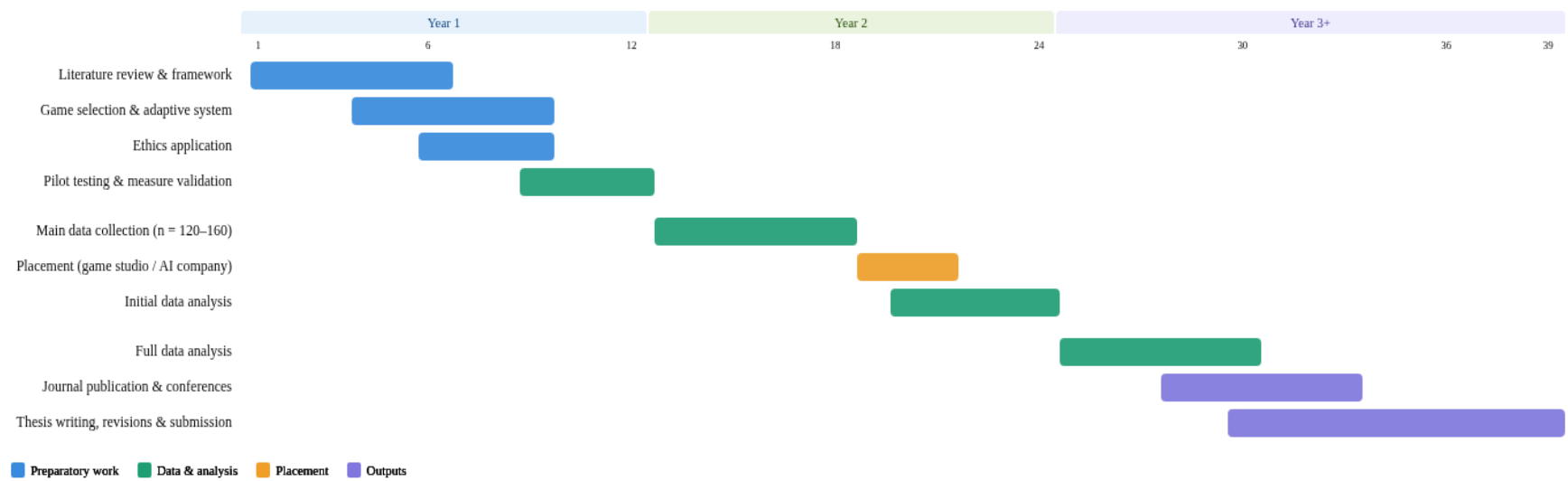


Figure 2. Proposed project timeline across the 3.5-year studentship. Overlapping bars indicate concurrent activities.

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